ABDELMONIEM OSAMA FATHALLAH MOHAMED ELHIFNAWY

SOFTWARE ENGINEER

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I'm a Software Engineer with over 8 years of experience designing and delivering complex software systems across Embedded Systems, Linux, Robotics, Android, and Enterprise Applications. Throughout my career, I've evolved from a hands-on engineer to a technical leader — driving architecture decisions, leading cross-functional teams, and mentoring developers to build scalable, maintainable, and reliable systems. I have a proven track record in system design, CI/CD automation, and diagnostic architectures, with a strong focus on improving developer efficiency, enforcing quality standards, and ensuring technical excellence across the stack.

WORK EXPERIENCE

Senior Software Engineer at GlobalLogic Poland

Mar/2023 - Present

GlobalLogic is a digital services company providing software product design and development services. It is an independent subsidiary of Hitachi Ltd.

Responsibilities

- Developed Android companion tooling used for internal automation, rapid diagnostic testing, and data collection.
- Designed Android helper apps to visualize and validate UDS/DIDs behaviors during ECU communication testing.
- Implemented Android companion apps for interacting with lab equipment (Test Rigs, Sensors, CAN logs).
- Led a cross-functional team of developers and testers to deliver AUTOSAR Diagnostic Stack features, ensuring architectural consistency and compliance with UDS and ISO standards.
- Designed and documented modular diagnostic architectures, integrating DIDs, DTCs, and communication protocols across multiple ECUs.
- Architected CI/CD pipelines using Jenkins to automate build, test, and deployment processes — improving release reliability and reducing manual interventions.
- Championed test automation using Robot Framework; defined testing standards, code review guidelines, and integration patterns.
- Mentored junior engineers on AUTOSAR, Jenkins automation, and diagnostic development best practices.
- Built internal developer tools and frameworks, including a VSCode extension to streamline task automation, tool access, and Test Rig connectivity.
- Created a map file analysis tool to validate linker section allocations and memory usage.
- Automated GreenHills IDE to capture and visualize OS task performance in real time, supporting in-depth performance analysis.
- Drove process improvements across teams through automation scripts, standardized workflows, and performance monitoring dashboards.
- Collaborated closely with architects and stakeholders to ensure feature roadmaps aligned with system design goals and project timelines.

Senior Embedded Software Engineer at Vehiclevo Egypt

May/2022 - Mar/2023

Vehiclevo is a German automotive software company founded in 2016, specializing in intelligent software architectures for ECUs, with a focus on EVs, infotainment, and autonomous driving. It operates globally, including a development center in Egypt, and is a recognized AUTOSAR partner.

Responsibilities

- Contributed to Android utility apps used for validating diagnostic configurations and performing automated test cycles.
- Developed Android data-collection helpers to support CAPL-based simulation testing.
- Built small Android companion apps to visualize simulation outputs and diagnostic responses.
- Developed advanced and sophisticated infotainment systems using Android, including Android Auto integration, real-time media playback, navigation, and vehicle status visualization.
- Implemented high-performance Android background services interacting with CAN bus and ECUs for real-time vehicle data and infotainment control.
- Optimized Android performance for embedded automotive environments, including multi-threading, gesture handling, and touch input responsiveness.
- Ensured seamless integration between Android components and AUTOSAR modules for consistent HMI behavior across vehicle systems.
- Led architectural configuration of AUTOSAR Basic Software using Capital VStar (CVI), ensuring consistent integration across projects.
- Designed simulation frameworks in Vector CANoe with CAPL for validating ECU communication logic.
- Introduced process automation for configuration consistency and reduced manual integration effort.
- Mentored new team members on AUTOSAR methodology, system configuration, and diagnostic design principles.
- Coordinated technical discussions with German teams to align architecture decisions and software design practices.

Embedded Software Engineer at Valeo

Oct/2018 - Apr/2022

Valeo is an automotive supplier and partner to automakers worldwide. It is a global leader in the design, production, and sale of components, integrated systems, and modules for the automotive industry.

Responsibilities

- Developed Android debugging apps to support testing of touch sensors and user-interaction systems.
- Created Android tools that visualized noise, signal levels, gesture detections, and firmware reaction timing.
- Collaborated with teams to integrate Android-based validation into HMI development workflows.
- Developed Android-based HMI components for infotainment touch displays, including media controls, notifications, and background services.
- Implemented multi-touch input handling and gesture recognition for steering wheel and dashboard interfaces.
- Integrated Android components with vehicle sensors and CAN bus for real-time feedback.
- Took ownership of steering systems architecture, designing modular AUTOSAR-compliant components for touch-based interfaces and light bands.
- Designed scalable embedded software frameworks for capacitive touch and HMI systems, reused across multiple projects and platforms.
- Defined coding and testing standards to improve maintainability and traceability across large automotive projects.
- Collaborated with cross-disciplinary teams to ensure robust system integration.
- Acted as a bridge between customer requirements and technical design, maintaining alignment using IBM DOORS and internal tools.
- Led multiple feature teams during critical project milestones, ensuring delivery on schedule and adherence to architectural guidelines.

Embedded Software Intern at Valeo

Mar/2018 - Jul/2018

Responsibilities

- Assisted in creating small Android experiments for sensor/gesture testing prototypes.
- Assisted in developing Android-based prototypes for infotainment HMI systems, including touch interfaces and basic multimedia playback.
- Assisted in the development of low level MCAL modules for CAN bus and touch sensors, ensuring system stability and performance.
- Assisted in the confgiguration of AUTOSAR Basic Software, ensuring consistent integration across projects.

Senior Software Engineer at KAN

Dec/2017 - Feb/2018

Responsibilities

- Architected Android apps and backend systems with Java2EE, SQL, and JSP, optimizing for performance and scalability.
- Developed Android front-ends for enterprise apps integrating with JavaEE backends.
- Built Android screens, services, network layers for internal business tools.
- Led small development teams to deliver client projects end-to-end, from requirements gathering to deployment.

Game Developer at Hive Studio

Dec/2017 - Feb/2018

Responsibilities

- Developed and shipped Android builds for VR/AR games using Unity.
- Optimized Android performance for 3D assets, rendering, and input handling.
- Created Android builds and mobile-optimized gameplay loops.

Android Developer at AL Hadeya For Arabic Software

Dec/2017 - Feb/2018

Responsibilities

- Built production ready Android apps for multiple clients.
- Developed UI, networking, background services, and database layers.
- Integrated translations, RTL support, and Arabic UI/UX patterns.
- Managing SQL databases and creating backend servers using Java2EE and JSP.

EDUCATION

Helwan University

Helwan, Egypt

Degrees

- Bachelor of Engineering

Sep/2012 - Sep/2017

California Polytechnic State University

California, USA

Degrees

- Training in Robotics

Jul/2016 - Aug/2016

PROJECTS

AlQuran

AlQuran is an app that allows you to listen to chapters of the Quran, voiced by different reciters. It also supports Android Auto. Features:

- Full Jetpack Compose UI with Material3 theming
- Streaming audio player with ExoPlayer
- Background playback with notifications and lockscreen controls
- Picture-in-Picture support, widgets, and local caching
- Arabic plural rules, RTL layout, dynamic colors
- Adaptive UI for different screen sizes and orientations
- Listen to chapters of the Quran, voiced by different reciters.
- Support for Android Auto.

Tools

Shell Scripting | *PowerShellScripting* | *VisualBASIC Scripting* | *Python* | *Kotlin* | *Jetpack Compose* | *Android* | *GitHub Worflow Actions*

Caffeinate

Caffeinate is an App that helps android developers to keep their phone's display awake without having to change the device settings. Features:

- It can be toggled from a tile in the quick settings, the place that holds the toggles for e.g. Wi-Fi and Bluetooth. Requires Android 7 or higher.
- It can be toggled from a button in the app itself
- It can be toggled from a widget in the home screen
- Uses android's WakeLock Class with either SCREEN_DIM_WAKE_LOCK or SCREEN_BRIGHT_WAKE_LOCK lock level based on user settings
- Timeout period is configurable
- Switch timeouts from the quick settings tile or the button in the app. clicking them, selects the next timeout and after 1 second the timeout will start, if the quick settings tile or the button are clicked after starting, the timeout will stop.
- Available timeouts are [30 seconds, 05 minutes, 10 minutes, 15 minutes, 30 minutes, 60 minutes, ∞]
- An option to enable the screen to dim while the WakeLock is acquired
- An option to enable holding the WakeLock if the screen is locked, so that the screen will keep on after unlocking, default behaviour is that the WakeLock is released when the screen is locked
- Multiple theming options [light, dark, system default and material you]

Tools

Shell Scripting | *Kotlin* | *Android* | *GitHub Worflow Actions*

CircularDurationView

A custom view for Android that displays a duration in a circular fashion. Features:

- Displays a duration in a circular format.
- The duration can be in seconds, minutes or hours.
- The progress indicators can be animated or not.
- The view is divided into three parts: hours, minutes and seconds.
- For each part, a progress indicator is displayed to show the progress of that part of the duration.
- A text view is displayed at the center of the view to show the duration in a readable format.

Tools

Shell Scripting | *Kotlin* | *Android* | *GitHub Worflow Actions*

Hands On Detection

Creating steering wheel hands on detection using segmented ADC segments and integrating with autonomous driving and applying gesture detection with which a user can play next/previous song or increase/decrease volume.

Steering Wheel Light Bands

Creating segmented steering wheel light bands and integrating with autonomous driving to indicate various states, show welcome light and display speed value through the light bands

Touch Display Based Steering Wheel Switches

Creating touch display based steering wheel switches with haptic feedback to show various information for the user such as currently playing song, caller info, cruise control info and more. Users can also interact with these displays to set various settings such as answering calls, changing music or setting cruise control speed.

Capacitive Touch Stack with Haptic Feedback

Creating in house capacitive touch stacks for use in capacitive touch based steering wheel switches, touch based displays or touchpads. The stack consisted of CAPS, CSBN, TSDS, SLID, TPAD, GDET and SENS modules which are Capacitive Sensing (low level layer which interfaces with the hardware and ADC peripherals), Baselining (baselining read ADC values over a period of time, applying filters and frequency hopping), Touch State Detection System (determining winning buttons in a group based touch steering wheel switches), Slider (detecting slider position in a touch based linear slider sensor with center of mass detection), Touch Pad (detecting multi-finger X, Y positions on a 2D touchpad with multiplexed NxM matrix), Gesture Detection (detecting 2D simple and complex gestures on a touchpad such as tapping, double tapping, one/two finger swipe directions, one/two finger scrolling, zooming in/out and more) and finally SENS which is a controller module for the previous modules that initiates and processes each of the stack's module in a pre-defined periodic task

eShifter

Creating applications for detecting/inducing transmission states in an electronic shift by wire shifter with priority-based processing for the P/R/N/D buttons and multi-push handling algorithms

Smart Home Automation System

a system where you can control your home's appliances from your smartphone while in the home's vicinity.

Tools

Android | AVR | Raspberry Pi | relays | motors | Bluetooth | Embedded C | Java | Python

Smart Home Security and Monitoring System

a system where you can monitor your home via a website from anywhere in the world, the system also works automatically to protect your home from hazardous events such as fire, water floods and glass breakage

Tools

Raspberry Pi | Arduino | relays | motors | solenoids | temperature / pressure sensors | Java | ArduinoC | Python

Autonomous Mobile Robot

a smart robot designed to play the game of capture the flag autonomously competing with other similar robots the, user can control the robot through his/her phone or a website manually or by giving it gps coordinates and letting the robot figure the best way to reach that point.

Tools

Android | Raspberry Pi | GPS sensors | IMU sensors | motors | Java | PHP | Python

AVR4L

AVR4L aims to deliver an easy to use and user-friendly Integrated Development Environment for developing Codes for microcontrollers based on Atmel's AVR families. AVR4L is a Creative coding / Integrated Development Environment for Linux operating systems intended for AVR beginners as well as professionals, it has many features from professional IDEs as well as the simplicity of editing

Tools

Linux | Bash | C | Java | avrdude | AVR-GCC | AVR

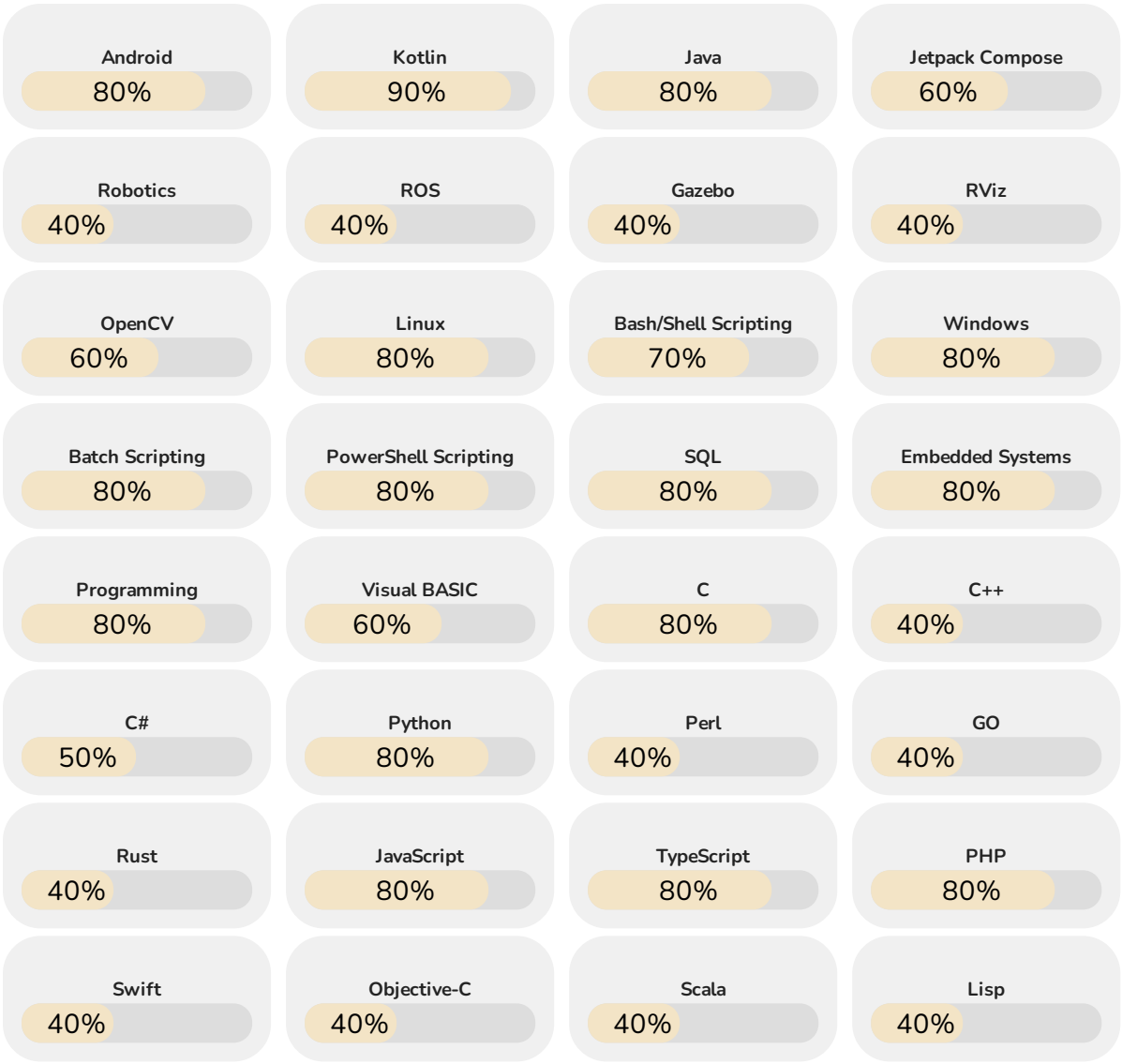
ALFA

ALFA is an Intelligent firefighting robotic system which helps extinguish fires and aid other human firefighters to accomplish this task quickly and swiftly with minimal losses, damages and most importantly saving lives. ALFA is an Acronym of Air and Land Firefighting Agents, it features a flying robot that is the brains of the system, scouts the area, scans and maps the environment, communicates with other robots and allocates tasks. It also features a group of ground bound robots that act as the muscles of the system, they are responsible for extinguishing the fire and navigating about the environment.

Tools

IMU | GPS Sensors | Motors | ROS | Gazebo | RViz | OpenCV | Computer Vision | AI | Linux | Bash | C++ | Python

SKILLS



VOLUNTEER EXPERIENCE

HR Member at Flickers

Sep/2015 - Aug/2015

Flickers is a student organization that aims to bridge the gap between school and job market in Egypt for the students of variant majors.

HR Executive Assistant at Flickers

Aug/2015 - Sep/2017

Flickers is a student organization that aims to bridge the gap between school and job market in Egypt for the students of variant majors.

Technical Mentor at NASA Space Apps - Cairo

Oct/2019 - Oct/2019

Participated as a mobile apps and ROS developers' mentor.

LANGUAGES

Arabic
Native speaker

English
Professional Working Proficiency